

Experiment :- 6

AIM :- Study and understand how SDLC, SRS, and Software Development Standards are integrated together in the BizFlow Inventory and Billing Management System project and how all previous experiments contribute to successful project development.

Introduction :-

Software development requires proper planning, requirement analysis, coding standards, testing, and documentation.

In this project, all previous experiments such as:

- SDLC Model
- Use Case Analysis
- Software Development Standards
- SRS Documentation

were combined together to build a complete and efficient Inventory and Billing Management System.

The integration of SDLC, SRS, and Standards helps in developing high-quality, reliable, secure, and maintainable software.

Integration of SDLC in the Project :-

SDLC Model Used :-

Agile Model

The Agile SDLC model was used because the project required continuous improvement, testing, and feature enhancement throughout development.

SDLC Phases Used in the Project :-

i) Requirement Gathering

Project requirements were collected:

- Product Management
- Inventory Tracking
- Bill Generation
- Invoice Generation
- Sales Analytics

- Bill History Management

Contribution:

Helped in understanding project goals clearly.

ii) Planning

Planning for:

- Software tools
- Database
- Modules
- Development Timeline

Contribution:

Provided proper project structure and roadmap.

iii) System Design

Designs prepared:

- Use Case Diagram
- Database Design
- System Architecture
- Module Design

Contribution:

Improved understanding of system workflow and implementation.

iv) Development

Modules developed:

- Authentication Module
- Product Catalog Module
- Inventory Management Module
- Billing Module
- Invoice Generation Module
- Analytics Dashboard Module

Contribution:

Converted design into working software.

v) Testing

Testing performed:

- Unit Testing
- Integration Testing
- System Testing

Contribution:

Reduced errors and improved software quality.

vi) Deployment

System deployed for practical use.

Contribution:

Made the project operational and accessible to users.

vii) Maintenance

System improvements and bug fixes were performed.

Contribution:

Improved project stability, reliability, and performance.

Integration of SRS in the Project :-

The SRS document helped define all project requirements clearly.

Functional Requirements Used

1. User Login
2. Product Management
3. Inventory Tracking
4. Bill Generation
5. Invoice PDF Generation
6. Bill History Management
7. Sales Analytics Dashboard

Non-Functional Requirements Used

1. High Performance
2. Security
3. Reliability
4. Scalability
5. User-Friendly Interface

Contribution of SRS

- Clear project understanding
- Better communication among stakeholders
- Reduced confusion during development
- Proper development planning

Integration of Software Development Standards :-

Software standards were followed during development to improve software quality.

Development Standards Used :-

i) Coding Standards

- Proper naming conventions
- Code formatting
- Meaningful comments

Contribution:

Improved code readability and maintenance.

ii) Design Standards

- UML Diagrams
- Database Design
- System Architecture

Contribution:

Improved system structure and planning.

iii) Testing Standards

- Unit Testing
- Integration Testing
- Error Handling

Contribution:

Improved software reliability and quality.

iv) Security Standards

- User Authentication
- Data Protection
- Secure Database Access

Contribution:

Improved system security and data protection.

Working Flow of Complete Project :-

1. User logs into the system
2. Product catalog is loaded from the database
3. User selects products and creates a bill
4. System calculates the bill amount automatically
5. Inventory is updated automatically after billing
6. Bill information is stored in the database
7. PDF invoice is generated
8. Sales analytics dashboard is updated
9. User can view bill history and reports

Benefits of Integrating SDLC, SRS, and Standards :-

i) Better Project Management

All project activities became organized and systematic.

ii) Improved Software Quality

Testing and standards reduced software errors.

iii) Easier Maintenance

Well-structured software became easy to update and enhance.

iv) Better Documentation

All project details were properly recorded and maintained.

v) Increased Reliability

System performance, security, and reliability improved significantly.

Technologies Used in BizFlow :-

Frontend:

- React.js
- Vite
- JavaScript
- Tailwind CSS

Backend & Database:

- Supabase
- PostgreSQL

Additional Features:

- PDF Invoice Generation
- Analytics Dashboard
- Inventory Automation

Conclusion :-

The successful integration of SDLC, SRS, and Software Development Standards helped in the development of BizFlow Inventory and Billing Management System. Proper planning, documentation, coding practices, and testing ensured that the software became reliable, maintainable, secure, and efficient for inventory and billing operations.

